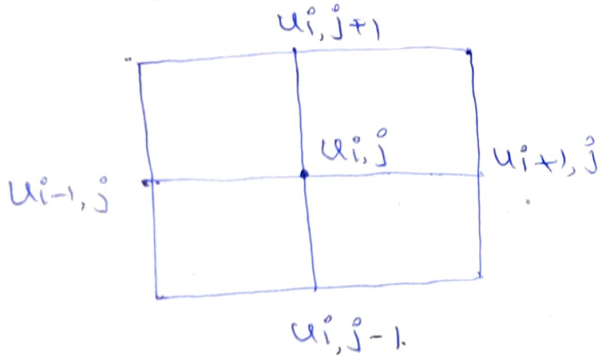
Solution of 2 dimensional Laplace equation (Elliptic eqn)

Suppose x & y are 2 independent variables & u is a real valued function of x & y. Then PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is known as 2 dimensional Laplace eqn.

Note: To solve this (1) we use standard 5-point formula (S.F) i.e.

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1}\right] \longrightarrow (1)$$

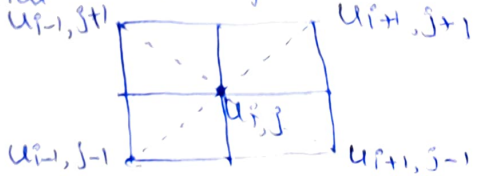
i.e $u_{i,j}$ at any interior mesh point is the average of its values at 4 neighbouring points to the left, right, above & below.



(2) We can use diagonal 5 point formula (D.F)

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j+1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j-1}\right] \longrightarrow (2)$$

i.e $u_{i,j}$ is the average of the values at 4 neighbouring diagonal mesh points.



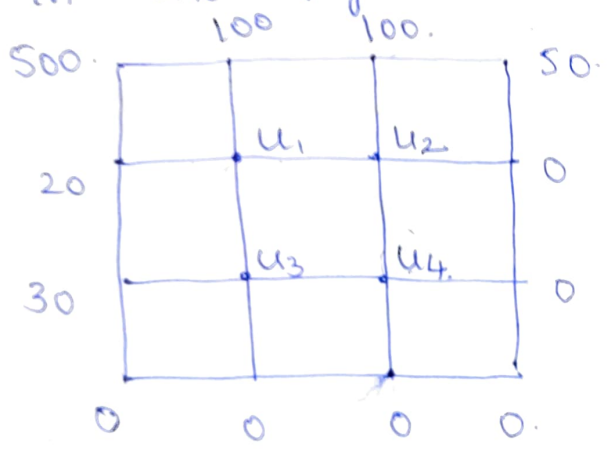
Sf is more accurate than D.F

(3) The accuracy of the value can be improved by successive application of the formula (3) we regard the obtained values of $u_{i,j}$ as the initial approximation denoted by $u_{i,j}^{(0)}$, then the first iteration is carried out by using the latest iterative value available on hand, where we have i.e

$$u_{i,j}^{(1)} = \frac{1}{4}\left[u_{i-1,j}^{(0)} + u_{i+1,j}^{(0)} + u_{i,j-1}^{(0)} + u_{i,j+1}^{(0)}\right]$$

It is repeated till we get the degree of accuracy. This process is called Liebmann's iteration process or Gauss-Seidel process.

① Solve $u_{xx} + u_{yy} = 0$ in the following square region with the boundary conditions as in the fig.



Apply standard 5 point formula

Let $u_4 = 0$.

$u_1 = \frac{1}{4}[500 + 30 + 100 + 0] \Rightarrow \boxed{u_1 = 157.5}$ [DF]

$u_2 = \frac{1}{4}[\overset{u_1}{157.5} + 100 + 0 + \overset{u_4}{0}] \Rightarrow \boxed{u_2 = 64.25}$ (SF)

$u_3 = \frac{1}{4}[30 + \overset{u_4}{0} + \overset{u_1}{157} + 0] \Rightarrow \boxed{u_3 = 46.75}$ (SF)

$\boxed{u_4 = 0}$ $\frac{1}{4}[u_3 + u_2 + 0 + 0] = \frac{1}{4}[46.75 + 64.25 + 0]$ [SF] $= 27.75$

Now $u_1^{(0)} = 157.5$, $u_2^{(0)} = 64.25$, $u_3^{(0)} = 46.75$

$u_4^{(0)} = 27.75$

Apply ~~diagonal~~ SF formula

$u_1 = \frac{1}{4}[20 + 100 + u_2 + u_3]$

$\Rightarrow 4u_1 - u_2 - u_3 = 120$.

$A = \frac{1}{4}(120 + B + C)$.

$u_2 = \frac{1}{4}[u_1 + 0 + 100 + u_4]$

$\Rightarrow 4u_2 - u_1 - u_4 = 100$.

$B = \frac{1}{4}(100 + A + D)$

$u_3 = \frac{1}{4}[30 + u_1 + u_4 + 0]$

$4u_3 - u_1 - u_4 = 30$.

$C = \frac{1}{4}(30 + u_1 + u_4)$

$A + D$

$u_4 = \frac{1}{4}(u_3 + u_2 + 0 + 0)$

$D = \frac{1}{4}(B + C)$

Insert eqns: $A = \frac{1}{4}(120 + B + C)$; $B = \frac{1}{4}(100 + A + D)$; $C = \frac{1}{4}(30 + A + D)$

$D = \frac{1}{4}(B + C)$

Insert

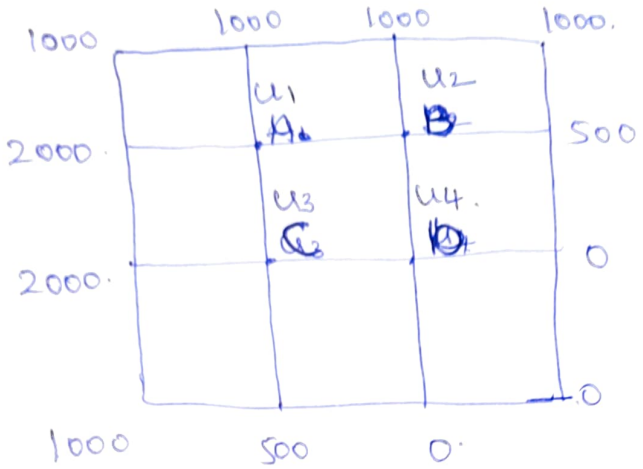
u_1	u_2	u_3	u_4
57.75	46.375	28.875	18.8125
48.8125	41.9063	24.4063	16.5781
46.5781	40.7891	23.2891	16.0195
46.0195	40.5098	23.0098	15.8799
45.8799	40.4399	22.9399	15.8450
45.845	40.4225	22.9225	15.8362
45.8362	40.4181	22.9181	15.8341
45.8341	40.4170	22.9170	15.8335
45.8335	40.4168	22.9168	15.8334
45.8334	40.4167	22.9167	15.8333

Since 9th & 10th iterations are exactly equal upto 3rd decimal places So stop

$\therefore$ $u_1 = 45.8334$ $u_2 = 40.4167$ $u_3 = 22.9167$

$u_4 = 15.8333$.

(2) Given the values of $u(x,y)$ on the boundary of the square in the fig given below, evaluate the fun $u(x,y)$ satisfying the Laplace eqn $\nabla^2 u = 0$ at the pivotal points of this fig.



To get the initial values of A, B, C & D. we assume u_4. $D = 0$. Then.

$$A = \frac{1}{4}(2000 + 1000 + 1000 + 0) \quad (DF)$$

$$\Rightarrow \boxed{A^{(0)} = 1000}$$

$$B = \frac{1}{4}(A + D + 1000 + 500)$$

$$B = \frac{1}{4}(1000 + 0 + 1000 + 500) \quad (SF)$$

$$\boxed{B^{(0)} = 625}$$

$$C = \frac{1}{4}(2000 + 500 + A + D) \quad (SF)$$

$$= \frac{1}{4}(2000 + 500 + 1000 + 0)$$

$$\boxed{C^{(0)} = 875}$$

$$D = \frac{1}{4}(C + B + 0 + 0)$$

$$= \frac{1}{4}(875 + 625 + 0) =$$

$$D = 375$$

$$\therefore \quad A^{(0)} = 1000, \quad B^{(0)} = 625, \quad C^{(0)} = 875, \quad D^{(0)} = 375$$

Now

$$A^{(1)} = \frac{1}{4}(2000 + 1000 + C + B)$$

$$A = 4^{-1}(3000 + B + C)$$

$$B^{(1)} = \frac{1}{4}(1000 + 500 + A + D)$$

$$B^{(1)} = 4^{-1}(1500 + A + D)$$

$$C^{(1)} = \frac{1}{4}(2000 + A + D + 500)$$

$$C^{(1)} = 4^{-1}(2500 + A + D)$$

$$D^{(1)} = \frac{1}{4}(B + C + 0)$$

$$\Rightarrow D = 4^{-1}(B + C).$$

Insert all the eqns in calculator.

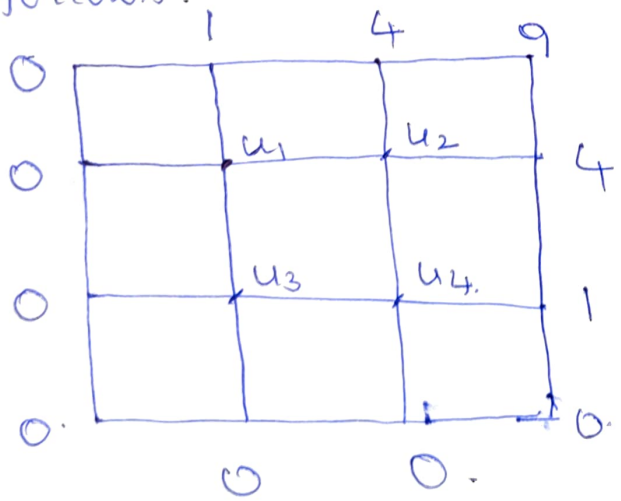
$A = 4^{-1}(3000 + B + C)$: $B = 4^{-1}(1500 + A + D)$: $C = 4^{-1}(2500 + A + D)$

$D = 4^{-1}(B + C)$

A	B	C	D
1125	750	1000	437.5
1187.5	781.25	1031.25	453.125
1203	789.0625	1039.0625	457.0313
1207.0313	791.0156	1041.0156	458.0078
1208.0078	791.5039	1041.5039	458.25
1208.252	791.62	1041.626	458.313
1208.3130	791.6565	1041.6565	458.3282

HW

③ Solve $\nabla^2 u = 0$ in the following square region with the boundary conditions as follows.



Let $u_4 = 0$.

$u_1 = \frac{1}{4}(0 + 4 + \tilde{u_4} + 0) \Rightarrow \boxed{u_1 = 1}$ (SF)

$$u_2 = \frac{1}{4}\left[u_1 + 4 + 4 + \overset{0}{u_4}\right] \text{ (DF)} = \frac{1}{4}[1+4+4] = \frac{9}{4} = 2.25$$

$$u_3 = \frac{1}{4}[0 + u_1 + u_4 + 0] \text{ (D.F)} = \frac{1}{4}[1+0] = \frac{1}{4} = 0.25$$

$$u_4 = \frac{1}{4}[u_3 + u_2 + 1 + 0] \text{ (D.F)} = \frac{1}{4}[0.25 + 2.25 + 1 + 0] = 0.875$$

$\therefore \quad u_1^{(0)} = 1, \quad u_2^{(0)} = 2.25, \quad u_3^{(0)} = 0.25, \quad u_4^{(0)} = 0.875.$

For I iteration SF.

$$u_1^{(1)} = \frac{1}{4}[0 + 1 + u_2^{(0)} + u_3^{(0)}]$$

$$u_2^{(1)} = \frac{1}{4}[u_1^{(1)} + 4 + 4 + u_3^{(0)}]$$

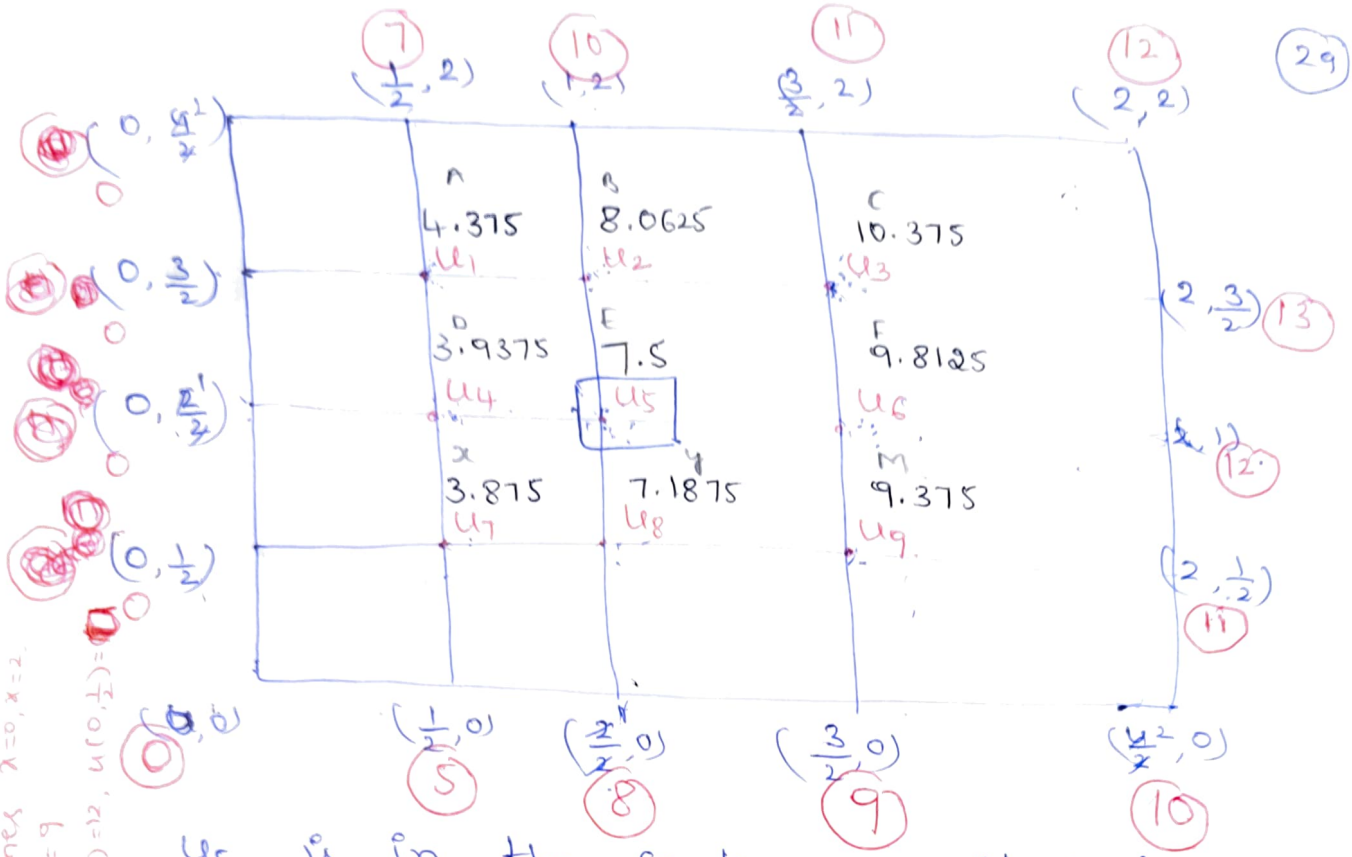
$$u_3^{(1)} = \frac{1}{4}[0 + u_1^{(1)} + u_4^{(0)} + 0]$$

$$u_4^{(1)} = \frac{1}{4}[u_3^{(1)} + u_2^{(1)} + 1 + 0]$$

u_1	u_2	u_3	u_4
0.875	0.2813	0.4375	0.1797
0.4297	0.2168	0.1523	0.0923
0.3423	0.1237	0.1086	0.0581
0.3081	0.1042	0.0915	0.0489
0.2989	0.0976	0.0870	0.0461
0.2961	0.0958	0.0856	0.0453
0.2953	0.0952	0.0852	0.0451
0.2951	0.0951	0.0850	0.045

Insert $A = 4^{-1}[1 + B + C]$; $B = 4^{-1}[A + C]$; $C = 4^{-1}[A + D]$; $D = 4^{-1}[B + C]$

Calc.



u_5 is in the centre, so start from u_5.

$$u_5 = \frac{1}{4}\left[11 + 10 + 8 + 12\right] \quad [SF] = 7.5$$

$$u_1 = \frac{1}{4}\left[0 + 11 + 10 + 7.5\right] \quad (DF) = 4.375$$

$$u_3 = \frac{1}{4}\left[12 + 12 + 7.5 + 10\right] \quad (DF) = 10.375$$

$$u_7 = \frac{1}{4}\left[11 + 8 + 7.5 + 11\right] \quad (DF) = 3.875$$

$$u_9 = \frac{1}{4}\left[10 + 8 + 12 + 7.5\right] \quad (DF) = 9.375$$

$$u_2 = \frac{1}{4}\left[4.375 + 10.375 + 10 + 7.5\right] \quad (SF) = 8.0625$$

$$u_4 = \frac{1}{4}\left[0 + 7.5 + 4.375 + 3.875\right] \quad (SF) = 3.9375$$

$$u_6 = \frac{1}{4}\left[7.5 + 12 + 10.375 + 9.375\right] \quad (SF) = 9.8125$$

(5) Solve $\nabla^2 u = 0$ following square mesh by the lines $x=0, x=2$, $y=0, y=2$ given that $u(0,0)=0$, $u(\frac{1}{2},0)=5$, $u(1,0)=8$ $u(\frac{3}{2},0)=9$, $u(2,0)=10$ $u(0,2)=10$ $u(\frac{1}{2},2)=7$, $u(1,2)=10$, $u(\frac{3}{2},2)=11$ $u(2,2)=12$, $u(0,\frac{1}{2})=0$, $u(0,1)=0$, $u(2,1)=12$ $u(2,\frac{1}{2})=11$, $u(0,\frac{3}{2})=0$, $u(2,\frac{3}{2})=13$. Carryout 3 iterations.

$$u_8 = \frac{1}{4}\left[3.875 + 9.375 + 8 + 7.5\right] \text{ (SF)} = 7.1875$$

I $u_1 = 4.375$, $u_2 = 8.0625$, $u_3 = 10.375$ $u_4 = 3.9375$

$u_5 = 7.5$ $u_6 = 9.8125$ $u_7 = 3.875$, $u_8 = 7.1875$

$u_9 = 9.375$

I iteration:

$$u_1^{(1)} = \frac{1}{4}\left[0 + u_2^{(0)} + 7 + u_4^{(0)}\right] = \frac{1}{4}\left[8.0625 + 7 + 3.9375\right]$$

$$u_1^{(1)} = \frac{1}{4}\left[$$

$$u_1^{(1)} = 7.5$$

$$u_2^{(1)} = \frac{1}{4}\left[u_1^{(0)} + u_3^{(0)} + 4 + u_5\right] = \frac{1}{4}\left[375 + 10.375 + 4 + 7.5\right] =$$

$$\boxed{u_2^{(1)} = 6.5625}$$

$$u_3^{(1)} = \frac{1}{4}\left[u_2^{(1)} + u_6^{(0)} + 15 + u_5\right] = \frac{1}{4}\left[6.5625 + 9.8125 + 15 + 7.5\right] \Rightarrow \boxed{u_3^{(1)} = 9.7188}$$

$$u_4^{(1)} = \frac{1}{4}\left[0 + u_1^{(1)} + 45 + u_7^{(0)}\right] = \frac{1}{4}\left[4.75 + 7.5 + 3.875\right]$$

$$\boxed{u_4^{(1)} = 4.0313}$$

$$u_5^{(1)} = \frac{1}{4}\left[u_4^{(1)} + u_6^{(0)} + u_2^{(1)} + u_8^{(0)}\right]$$

$$= \frac{1}{4}\left[4.0313 + 9.8125 + 6.5625 + 7.1875\right]$$

$$\boxed{u_5^{(1)} = 6.8985}$$

$$u_6^{(1)} = \frac{1}{4}\left[u_5^{(1)} + u_3^{(1)} + u_9^{(0)} + 14\right] = \frac{1}{4}\left[6.8985 + 9.7188 + 9.375 + 14\right]$$

$$\boxed{u_6^{(1)} = 9.9981}$$

$$u_7^{(1)} = \frac{1}{4}\left[0 + u_8^{(0)} + u_4^{(1)} + 3\right] = \frac{1}{4}\left[0 + 7.1875 + 4.0313 + 3\right]$$

$$\boxed{u_7^{(1)} = 3.5547}$$

$$u_8^{(1)} = \frac{1}{4}\left[6 + u_5^{(1)} + u_9^{(0)} + u_7^{(1)}\right] = \frac{1}{4}\left[6 + 6.8985 + 9.375 + 3.5547\right]$$

$$\boxed{u_8^{(1)} = 6.4571}$$

$$u_9^{(1)} = \frac{1}{4}\left[u_8^{(1)} + 13 + u_6^{(1)} + 9\right] = \frac{1}{4}\left[6.4571 + 13 + 9.9981 + 9\right]$$

$$\boxed{u_9^{(1)} = 9.6138}$$

$u_1^{(1)} = 4.75$, $u_2^{(1)} = 6.5625$, $u_3^{(1)} = 9.7188$
$u_4^{(1)} = 4.0313$, $u_5^{(1)} = 6.8985$, $u_6^{(1)} = 9.9981$
$u_7^{(1)} = 3.5547$, $u_8^{(1)} = 6.4571$, $u_9^{(1)} = 9.6138$.

Similarly we can find 2nd iteration by taking latest value, by applying standard five point formula

$$u_1^{(2)} = \frac{1}{4}\left[0 + 7 + \overset{6.5625}{\cancel{8.065}} + 3.9375\right] \quad u_1^{(2)} = 5.1484.$$

$$u_2^{(2)} = \frac{1}{4}\left[u_1 + 10 + u_3 + u_5\right] = \frac{1}{4}\left[5.1484 + 10 + 9.7188 + 6.8985\right]$$

$$u_2^{(2)} = 7.9414.$$

$$u_3^{(2)} = \frac{1}{4}\left[u_2 + 11 + 13 + u_6\right] = \frac{1}{4}\left[7.9414 + 24 + 9.9981\right]$$

$$u_3^{(2)} = 10.4849$$

$$u_4^{(2)} = \frac{1}{4}\left[0 + u_1 + u_5 + u_7\right] = \frac{1}{4}\left[5.1484 + 6.8985 + 3.5547\right)$$

$$u_4^{(2)} = 3.9004.$$

$$u_5^{(2)} = \frac{1}{4}\left[u_2 + u_4 + u_6 + u_8\right] = \frac{1}{4}\left[7.9414 + 3.9004 + 9.9981 + 6.4571\right]$$

$$u_5^{(2)} = 7.0743$$

$$u_6^{(2)} = \frac{1}{4}\left[u_5 + u_3 + 12 + u_9\right] = \frac{1}{4}\left[7.0743 + 10.4849 + 12 + 9.6138\right]$$

$$u_6^{(2)} = 9.7933$$

$$u_7^{(2)} = \frac{1}{4}\left[0 + 5 + u_4 + u_8\right] = \frac{1}{4}\left[5 + 3.9004 + 6.4571\right]$$

$$u_7^{(2)} = 3.8394$$

$$u_8^{(2)} = \frac{1}{4}\left[u_5 + u_7 + u_9 + 8\right] = \frac{1}{4}\left[7.0743 + 3.8394 + 9.6138 + 8\right]$$

$$u_8^{(2)} = 7.1319.$$

$$u_9^{(2)} = \frac{1}{4}\left[u_6 + u_8 + 11 + 9\right] = \frac{1}{4}\left[9.7933 + 7.1319 + 20\right]$$

$$u_9^{(2)} = 9.2313$$

⑥ Solve $\nabla^2 u = 0$ in the square region bounded by the coordinate axes & lines $x=4$, $y=4$ with the boundary conditions given by analytical expressions

① $u(0,y) = 0$ for $0 \le y \le 4$

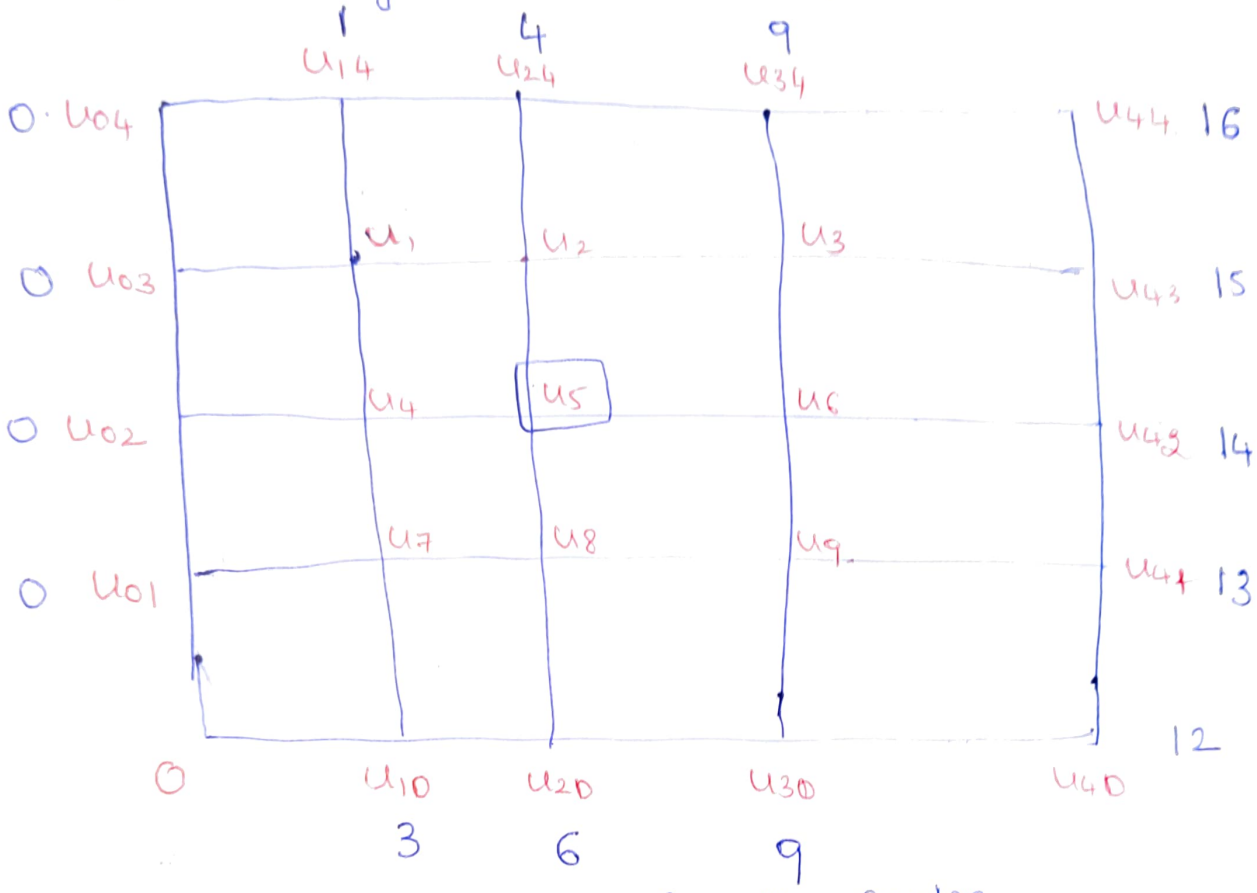
② $u(4,y) = 12 + y$ $\quad 0 \le y \le 4$

③ $u(x,0) = 3x$ $\quad 0 \le x \le 4$.

④ $u(x,4) = x^2$ $\quad 0 \le x \le 4$

Perform 2 iterations

Soln: We have $\nabla^2 u = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, divide the square region into 16 square meshes.



u_5 is exactly in the centre.

$$u_5 = \frac{1}{4}[0 + 14 + 6 + 4] = 6 \text{ (SF)}$$

$$u_1 = \frac{1}{4}[0 + 4 + u_5 + 0] = \frac{16}{4} = 2.5 \text{ (D.F.)}$$

$$u_3 = \frac{1}{4}[6 + 4 + 16 + 14] \text{ DF} = 10$$

$$u_7 = \frac{1}{4}[0 + 6 + 6 + 0] \text{ D.F} = 3$$

$$u_9 = \frac{1}{4}[6 + 12 + 14 + 6] \text{ (D.F)} = 9.5$$

$$u_2 = \frac{1}{4}[2.5 + 10 + 4 + 6] \text{ (SF)} = 5.875$$

$$u_4 = \frac{1}{4}[0 + 6 + 2.5 + 3) \text{ (SF)} = 2.875$$

$u_6 = \frac{1}{4}(6+14+10+9.5)$ (SF) $= 9.875$

$u_8 = \frac{1}{4}(3+6+9.5+6)$ (SF) $= 6.125$

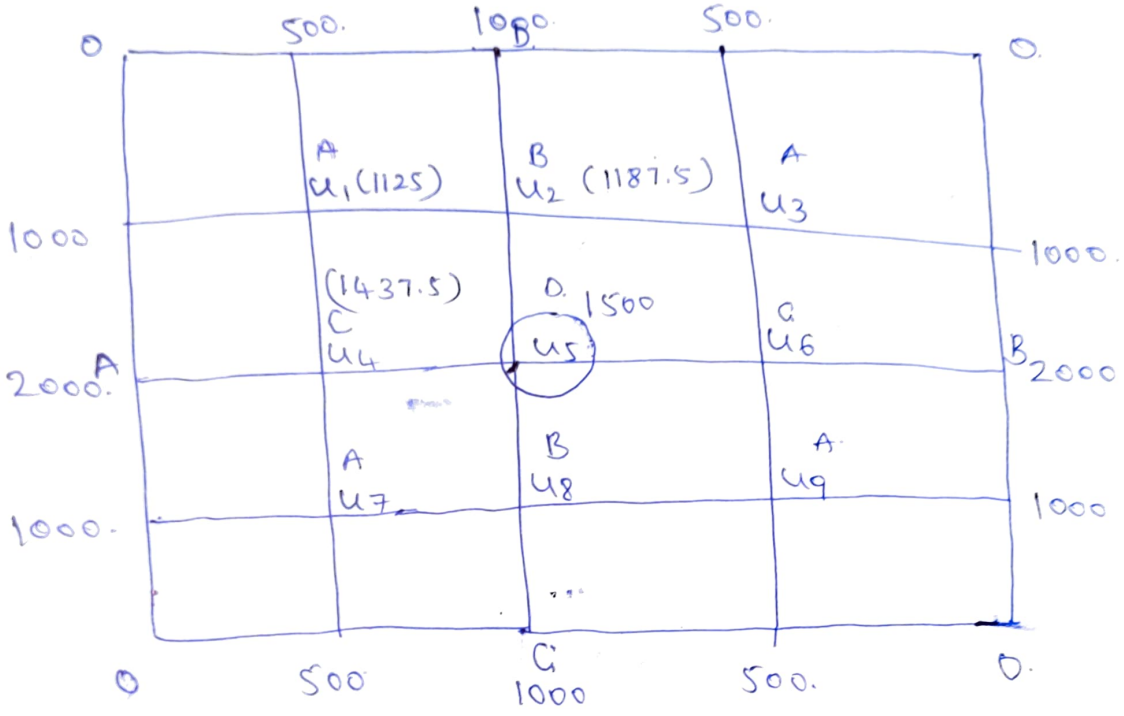
Now $u_1^{(0)} = 1.75$ $u_2^{(0)} = 5.4375$, $u_3^{(0)} = 10$, $u_4^{(0)} = 2.6875$

$u_5^{(0)} = 6$ $u_6^{(0)} = 9.875$ $u_7^{(0)} = 3$, $u_8^{(0)} = 6.125$

$u_9^{(0)} = 9.5$

Apply S.F for I iteration.

① Solve the elliptic eqn $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values as shown in the fig



The boundary value of u are symmetrical about AB $\therefore u_1 = u_7$, $u_2 = u_8$ $u_3 = u_9$

CD : $u_1 = u_3$, $u_4 = u_6$, $u_7 = u_9$

i.e $u_1 = u_7 = u_3 = u_9$

We need to find u_1, u_2, u_4 & u_5

D. $u_5 = \frac{1}{4}[0+0+0+1000+2000+2000+1000]$ [SF] $= 1500$

A $u_1 = \frac{1}{4}(0+1000+1500+2000)$ [dF) $= 1125$

B $u_2 = \frac{1}{4}[u_1+1000+u_3+u_5]$ [SF] $u_2 = 1187.5$

C $u_4 = \frac{1}{4}[u_1+1500+2000+u_7]$ (SF) $= \frac{1}{4}[1125+1500+2000+1125]$

$u_4 = 1437.5$, $u_5 = D.$

Let $u_1 = A = u_3 = u_7 = u_9$, $u_2 = B = u_8$, $u_4 = C = u_6$,

$$u_1 = \frac{1}{4}[1000 + 500 + u_2 + u_4]$$

$$u_2 = \frac{1}{4}[1000 + u_1 + u_3 + u_5]$$

$$u_4 = \frac{1}{4}[2000 + u_1 + u_5 + u_7]$$

$$u_5 = \frac{1}{4}[u_2 + u_4 + u_6 + u_8]$$

Insert

$A = 4^{-1}[1500 + B + C]$: $B = 4^{-1}[1000 + 2A + D]$:

$C = 4^{-1}[2000 + 2A + D]$: $D = 4^{-1}[2B + 2C]$

u_1	u_2	u_4	u_5
1031.25	1140.625	1390.625	1265.625
1007.8125	1070.3125	1320.3125	1195.3125
972.6563	1035.1563	1285.1563	1160.1563
955.0781	1017.5781	1267.5781	1142.5781
946.2891	1008.7891	1258.7891	1133.7891
941.8945	1004.3945	1254.3945	1129.3945
939.6973	1002.1973	1252.1973	1127.1973
938.5986	1001.0986	1251.0986	1126.0986
938.0493	1000.5493	1250.5493	1125.5493
937.7747	1000.2747	1250.2747	1125.2747
937.6373	1000.1373	1250.1373	1125.1373
937.5687	1000.0687	1250.0687	1125.0687